

Relatividad especial

$$E = \sqrt{m^2 + |\vec{p}|^2} \quad \vec{p} = E \frac{\vec{u}}{c^2}$$

$$(p_i^\mu)^2 = (p_f^\mu)^2 \quad E = m\gamma \quad \tau = T/\gamma$$

$$\begin{cases} A^{0'} = \gamma(A^0 - vA^x) \\ A^{x'} = \gamma(A^x - vA^0) \\ A^{y'} = A^y; \quad A^{z'} = A^z \end{cases}$$

Isospin

Teniendo en cuenta: $\hat{I}_1, \hat{I}_2, \hat{I}_3 \in SU(2)$

$$[\hat{I}_i, \hat{I}_j] = i\epsilon_{ijk}\hat{I}_k \quad ; \hat{I}_\pm = \hat{I}_1 \pm i\hat{I}_2$$

$$\hat{I}^2 = \hat{I}_1^2 + \hat{I}_2^2 + \hat{I}_3^2$$

$$\mu_{jm} = \mu_j$$

$$\langle j', m' | \hat{U} | j, m \rangle = \delta_{jj'} \delta_{mm'} M_j$$

$$\langle j, m | \hat{U} | j, m \rangle = \langle j, m' | \hat{U} | j, m' \rangle$$

Postulado: $[\hat{H}_F, \hat{I}_j] = 0$

Modelo del quark

$$\left[\frac{\lambda_1}{2}, \frac{\lambda_2}{2} \right] = i \frac{\lambda_3}{2}$$

$$\left[\frac{\lambda_3}{2}, \frac{\lambda_5}{2} \right] = \frac{i}{4} (\lambda_3 + \sqrt{3}\lambda_8)$$

$$\left[\frac{\lambda_6}{2}, \frac{\lambda_7}{2} \right] = \frac{i}{4} (-\lambda_3 + \sqrt{3}\lambda_8)$$

$$\hat{I}_\pm = \frac{\lambda_1 \pm i\lambda_2}{2} \quad \hat{V}_\pm = \frac{\lambda_4 \pm i\lambda_5}{2}$$

$$\hat{U}_\pm = \frac{\lambda_6 \pm i\lambda_7}{2} \quad Q = I_3 + \frac{1}{2}Y$$

El **sabor** debe ser un singlete de $SU(3)_{color}$.

Para el **antifundamental** todas invierten y agregan signo.

Momento magnético

$$\hat{\mu} = \frac{q}{2m} \hat{\sigma} \quad (\text{Sigma es para el spin})$$

$$\hat{\mu}_{z,i} = Q_i \frac{e}{2m_i} \hat{\sigma}_3 \rightarrow i = u, d, s$$

Dirac y KG

$$(\partial^\mu \partial_\mu + m^2)\phi = 0 \quad (i\gamma^\mu \partial_\mu - m)\psi = 0$$

$$(k \cdot \vec{\sigma})^{-1} = \frac{k \cdot \vec{\sigma}}{|\vec{k}|^2}$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$$

$$(\gamma^0)^\dagger = \gamma^0 \quad (\gamma^i)^\dagger = -\gamma^i$$

Transformaciones

$$S(\omega) = e^{-\frac{i}{2}\omega_{\alpha\beta}\Sigma^{\alpha\beta}} \rightarrow \Sigma^{\alpha\beta} = \frac{1}{4}[\gamma^\alpha, \gamma^\beta]$$

Helicidad

$$h = \vec{s} \cdot \hat{\vec{p}} \quad " = " \frac{1}{2} \frac{\vec{\Sigma} \cdot \vec{p}}{|\vec{p}|} \quad [H, h] = 0$$

Quiralidad

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

$$[\gamma^5, S(\Lambda)] = 0; \quad [H, \gamma^5] \neq 0$$

$$\gamma^5 \approx 2h$$

$$(\gamma^5)^\dagger = \gamma^5; \quad (\gamma^5)^2 = 1$$

$$\{\gamma^5, \gamma^\mu\} = 0$$

$$\hat{P}_R = \frac{1}{2}(1 + \gamma^5); \quad \hat{P}_L = \frac{1}{2}(1 - \gamma^5)$$

Lagrange

$$\partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \right) - \frac{\partial L}{\partial \phi} = 0$$

$$\frac{\partial(\partial_\mu \phi)}{\partial(\partial_\nu \phi)} = \delta_\mu^\nu \quad \frac{\partial(\partial^\mu \phi)}{\partial(\partial_\nu \phi)} = \eta^{\mu\nu} \quad \frac{\partial(\partial_\mu A_\nu)}{\partial(\partial_\alpha A_\beta)} = \delta_\mu^\alpha \delta_\nu^\beta$$

Noether

Supongamos $L = L(\phi_1, \dots, \phi_N, \partial_\mu \phi_1, \dots, \partial_\mu \phi_N)$, aplicando transformación infinitesimal:

$$L' = L + \partial_\mu K^\mu$$

$$\Rightarrow J^\mu = \sum_i \frac{\partial L}{\partial(\partial_\mu \phi_i)} \Delta \phi_i + \partial_\mu K^\mu$$

$$\partial_\mu J^\mu = 0 \quad (\text{Bajo ecs de mto})$$